HW12 Part A :: Image Segmentation

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Setup

In [9]:
```
import numpy as np
import pandas as pd
import cv2
import matplotlib.pyplot as plt
from sklearn.preprocessing import MinMaxScaler
from sklearn.cluster import KMeans, MeanShift, estimate_bandwidth
import urllib.request
from pathlib import Path
import warnings

warnings.filterwarnings("ignore")

print("OpenCV:", cv2.__version__)
```

```
OpenCV: 4.10.0
```

Load the image

Download a sample colour image and place it in the repository root as required by the assignment.

In [10]:
```
REPO_ROOT = Path("../..")  # HW12/Part_A -> repo root
IMG_NAME = "segmentation_sample.jpg"
IMG_PATH = REPO_ROOT / IMG_NAME

if not IMG_PATH.exists():
    url = "https://raw.githubusercontent.com/opencv/opencv/master/samples/data
    req = urllib.request.Request(url, headers={'User-Agent': 'Mozilla/5.0'})
    with urllib.request.urlopen(req) as response, open(IMG_PATH, 'wb') as out_
        out_file.write(response.read())
    print(f"Downloaded sample image to {IMG_PATH.resolve()}")
else:
    print(f"Image already exists at {IMG_PATH.resolve()}")

image = plt.imread(str(IMG_PATH))
if image.max() <= 1.0:
    image = (image * 255).astype(np.uint8)
if image.ndim == 2:
    image = cv2.cvtColor(image, cv2.COLOR_GRAY2RGB)
elif image.shape[2] == 4:
    image = cv2.cvtColor(image, cv2.COLOR_RGBA2RGB)
```

```python
height, width, channels = image.shape

plt.figure(dpi=150)
plt.title("Original Image")
plt.imshow(image)
plt.axis("off")
plt.show()

print(f"Image shape: {image.shape}")
```

```
Image already exists at C:\Users\Adam\Desktop\projects_and_code\School\COSC637
3\HWs\segmentation_sample.jpg
```

Original Image

```
Image shape: (480, 512, 3)
```

Question 1

Please detail the differences between unsupervised image segmentation methods and supervised semantic image segmentation methods.

Aspect	Unsupervised Segmentation	Supervised Semantic Segmentation
Training data	No labelled data required	Requires large pixel-level annotated datasets
Learning paradigm	Groups pixels by similarity (colour, texture, spatial proximity) without any prior knowledge of object classes	Learns a mapping from pixels to predefined class labels using ground-truth masks
Output	Clusters / regions with no semantic meaning - segments are just groups of similar pixels	A label map where every pixel is assigned a meaningful class (e.g., *car*, *road*, *sky*)
Typical algorithms	K-Means, Mean-Shift, Gaussian Mixture Models, spectral clustering, watershed	FCN, U-Net, DeepLab, SegFormer, Mask R-CNN (instance)
Generalization	Works on any image without retraining; parameters (e.g., *k*) are set manually	Generalizes to unseen images of the same domain but requires retraining for new domains
Computational cost	Generally lighter at inference (no deep network)	Heavier - requires GPU for training and often for inference
Evaluation	Hard to evaluate objectively (no ground truth); often visual inspection	Standard metrics: mIoU, pixel accuracy, Dice coefficient

Question 2

Name well-known methods from each of the above categories.

Unsupervised image segmentation methods:

- K-Means clustering
- Mean-Shift clustering
- Gaussian Mixture Models (GMM / Expectation-Maximisation)
- Spectral clustering
- Watershed algorithm
- SLIC Superpixels
- Otsu's thresholding
- Graph-cut / Normalized Cuts

Supervised semantic segmentation methods:

- Fully Convolutional Network (FCN) - Long et al., 2015
- U-Net - Ronneberger et al., 2015
- DeepLab (v1-v3+) - Chen et al.
- SegFormer - Xie et al., 2021
- PSPNet (Pyramid Scene Parsing Network)
- Mask R-CNN (instance segmentation)
- HRNet (High-Resolution Network)

Question 3 - K-Means Image Segmentation

Build feature vectors

Following the reference article [1], we create two representations of the image pixels:

- `df_1` - RGB values *plus* pixel (col, row) location
- `df_2` - RGB values only

In [11]:
```
index = pd.MultiIndex.from_product(
    (*map(range, image.shape[:2]), ("r", "g", "b")),
    names=("row", "col", None),
)
df_1 = pd.Series(image.flatten(), index=index)
df_1 = df_1.unstack()
df_1 = df_1.reset_index().reindex(columns=["col", "row", "r", "g", "b"])

df_2 = df_1[["r", "g", "b"]]

print("df_1 (RGB + location):")
display(df_1.head())
print("\ndf_2 (RGB only):")
display(df_2.head())
```

df_1 (RGB + location):

	col	row	r	g	b
0	0	0	130	62	13
1	1	0	131	63	14
2	2	0	134	64	15
3	3	0	134	62	14
4	4	0	134	62	14

```
df_2 (RGB only):
```

	r	g	b
0	130	62	13
1	131	63	14
2	134	64	15
3	134	62	14
4	134	62	14

Normalize to 0-1

Pixel values are 0-255 while spatial coordinates can be much larger.
We normalize every feature to [0, 1] with `MinMaxScaler` so Euclidean distance treats all dimensions equally.

In [12]:
```
nd_1 = MinMaxScaler(feature_range=(0, 1)).fit_transform(df_1)
nd_2 = MinMaxScaler(feature_range=(0, 1)).fit_transform(df_2)

print(f"nd_1 shape: {nd_1.shape}  (col, row, r, g, b)")
print(f"nd_2 shape: {nd_2.shape}  (r, g, b)")
```

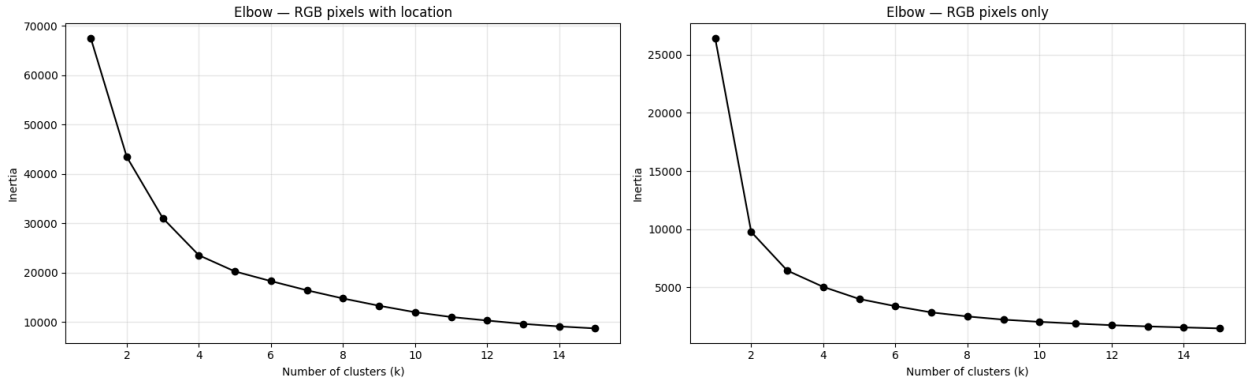
```
nd_1 shape: (245760, 5)  (col, row, r, g, b)
nd_2 shape: (245760, 3)  (r, g, b)
```

Elbow method - optimal *k*

Run K-Means for a range of *k* values on both feature vectors and plot inertia.

In [13]:
```
K_RANGE = range(1, 16)

inertias_1 = [KMeans(n_clusters=k, n_init=10, random_state=42).fit(nd_1).inert
inertias_2 = [KMeans(n_clusters=k, n_init=10, random_state=42).fit(nd_2).inert

fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(16, 5))

ax1.plot(list(K_RANGE), inertias_1, "o-", color="black")
ax1.set_xlabel("Number of clusters (k)")
ax1.set_ylabel("Inertia")
ax1.set_title("Elbow - RGB pixels with location")
ax1.grid(True, alpha=0.3)

ax2.plot(list(K_RANGE), inertias_2, "o-", color="black")
ax2.set_xlabel("Number of clusters (k)")
ax2.set_ylabel("Inertia")
ax2.set_title("Elbow - RGB pixels only")
ax2.grid(True, alpha=0.3)
```

```
plt.tight_layout()
plt.show()
```



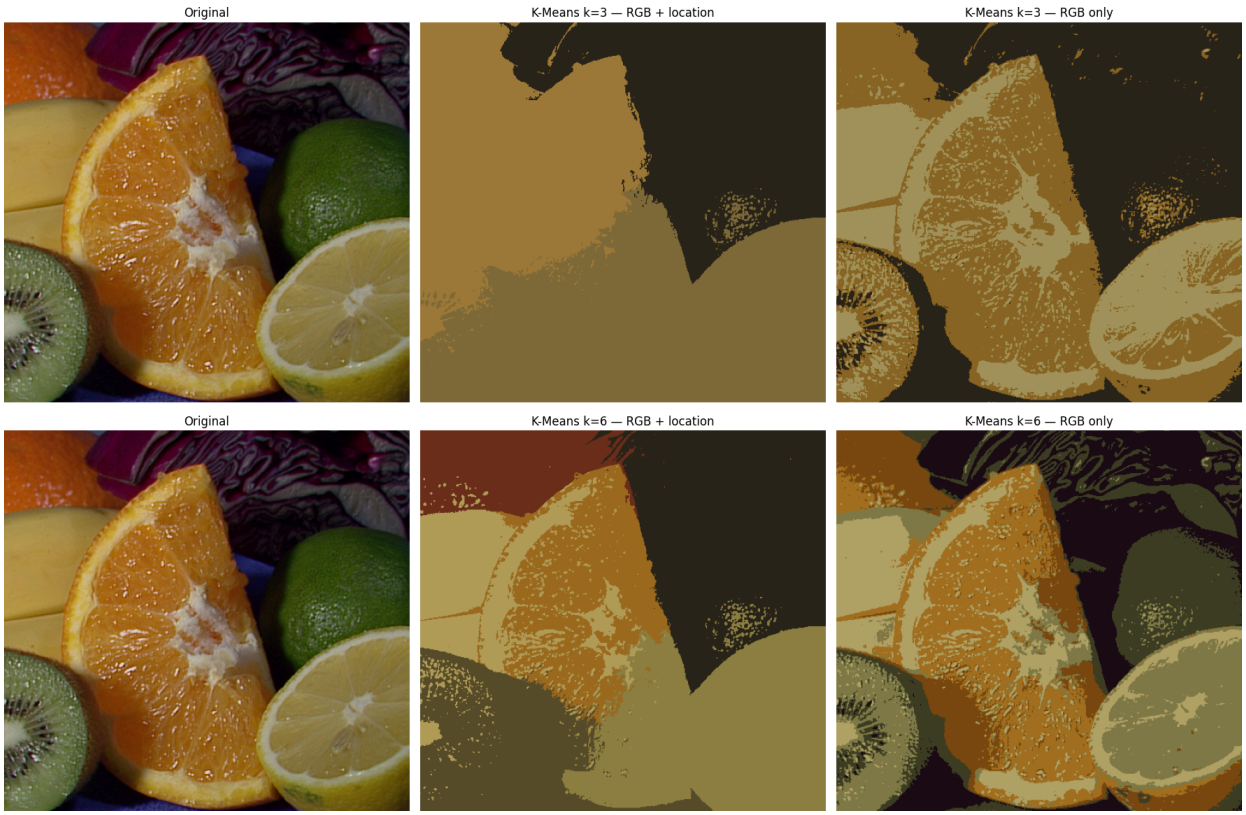
The elbow is generally more visible in the RGB-only plot because the feature space is lower-dimensional.
The feature vector that includes pixel location descends more smoothly, making the elbow harder to pinpoint - but it produces spatially coherent clusters.

Apply K-Means with two different *k* values

Based on the elbow plots, we pick two values of *k* and visualise the segmentation for both feature vectors.

In [14]:
```
def recreate_image(cluster_centers, labels, w, h):
    """Reconstruct an image from cluster centres and per-pixel labels."""
    return np.clip(cluster_centers[labels].reshape(h, w, 3), 0, 255).astype(np


k_values = [3, 6]

fig, axes = plt.subplots(len(k_values), 3, figsize=(18, 6 * len(k_values)))

for row, k in enumerate(k_values):
    # --- with location ---
    km1 = KMeans(n_clusters=k, n_init=10, random_state=42).fit(nd_1)
    rgb_centres_1 = km1.cluster_centers_[:, 2:] * 255  # last 3 cols are r,g,b
    seg1 = recreate_image(rgb_centres_1, km1.labels_, width, height)

    # --- RGB only ---
    km2 = KMeans(n_clusters=k, n_init=10, random_state=42).fit(nd_2)
    rgb_centres_2 = km2.cluster_centers_ * 255
    seg2 = recreate_image(rgb_centres_2, km2.labels_, width, height)

    axes[row, 0].imshow(image)
    axes[row, 0].set_title("Original")
    axes[row, 0].axis("off")

    axes[row, 1].imshow(seg1)
    axes[row, 1].set_title(f"K-Means k={k} - RGB + location")
```

```
    axes[row, 1].axis("off")

    axes[row, 2].imshow(seg2)
    axes[row, 2].set_title(f"K-Means k={k} - RGB only")
    axes[row, 2].axis("off")

plt.tight_layout()
plt.show()
```



Including pixel location in the feature vector produces spatially coherent clusters - nearby pixels of similar colour are grouped together.
The RGB-only vector segments purely by colour, so distant pixels that happen to share a colour end up in the same cluster.

Question 3.2 - Mean-Shift Clustering

Apply the Mean-Shift algorithm to the same image.
Mean-Shift does not require specifying the number of clusters - it discovers them automatically via kernel density estimation.
The critical parameter is the bandwidth, estimated here with `estimate_bandwidth`.

In [15]:
```
bandwidth_1 = estimate_bandwidth(nd_1, quantile=0.06, n_samples=1000)
```

```
bandwidth_2 = estimate_bandwidth(nd_2, quantile=0.06, n_samples=1000)

print(f"Bandwidth (with location):  {bandwidth_1:.4f}")
print(f"Bandwidth (RGB only):       {bandwidth_2:.4f}")
```

```
Bandwidth (with location):  0.2696
Bandwidth (RGB only):       0.1051
```

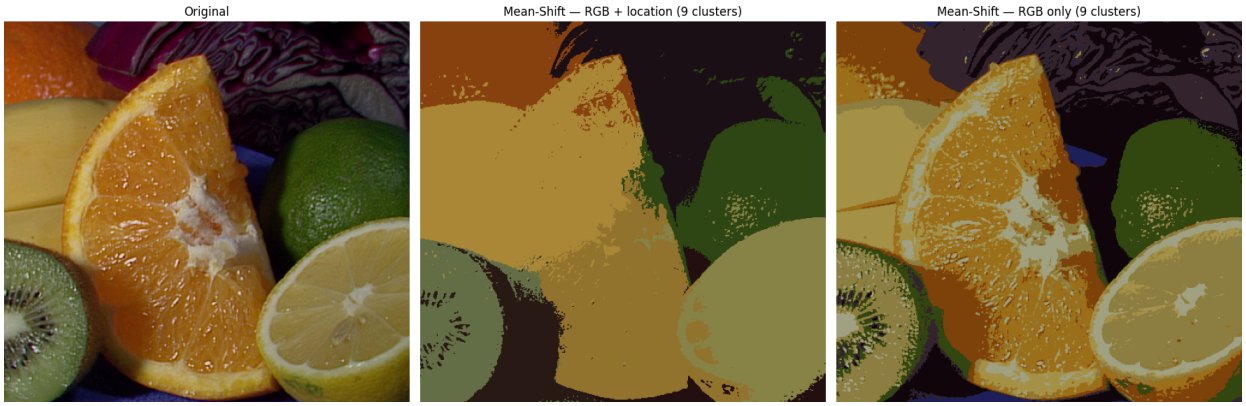
In [16]:

```
ms_1 = MeanShift(bandwidth=bandwidth_1, bin_seeding=True, cluster_all=True).fi
ms_2 = MeanShift(bandwidth=bandwidth_2, bin_seeding=True, cluster_all=True).fi

print(f"Mean-Shift (with location): {len(ms_1.cluster_centers_)} clusters")
print(f"Mean-Shift (RGB only):      {len(ms_2.cluster_centers_)} clusters")
```

```
Mean-Shift (with location): 9 clusters
Mean-Shift (RGB only):      9 clusters
```

In [17]:

```
rgb_centres_ms1 = ms_1.cluster_centers_[:, 2:] * 255
seg_ms1 = recreate_image(rgb_centres_ms1, ms_1.labels_, width, height)

rgb_centres_ms2 = ms_2.cluster_centers_ * 255
seg_ms2 = recreate_image(rgb_centres_ms2, ms_2.labels_, width, height)

fig, axes = plt.subplots(1, 3, figsize=(18, 6))

axes[0].imshow(image)
axes[0].set_title("Original")
axes[0].axis("off")

axes[1].imshow(seg_ms1)
axes[1].set_title(f"Mean-Shift - RGB + location ({len(ms_1.cluster_centers_)}
axes[1].axis("off")

axes[2].imshow(seg_ms2)
axes[2].set_title(f"Mean-Shift - RGB only ({len(ms_2.cluster_centers_)} cluste
axes[2].axis("off")

plt.tight_layout()
plt.show()
```



The feature vector that includes location information produces better spatial

grouping - nearby similar-colour regions are merged, which is more useful for tasks like identifying landing areas or road surfaces.
The RGB-only vector can over-segment because it ignores spatial proximity.

Question 3.3 - Is Image Thresholding a Form of Unsupervised Image Segmentation?

Yes. Image thresholding (e.g., Otsu's method, adaptive thresholding) is one of the simplest forms of unsupervised image segmentation.

- It partitions pixels into foreground and background (or multiple regions for multi-level thresholding) based solely on intensity values - no labelled training data is required.
- Otsu's method in particular automatically selects the threshold that minimises intra-class variance, which is analogous to how K-Means (with k = 2) partitions data.
- The key distinction from clustering-based methods is that thresholding typically operates on a 1-D histogram (intensity) rather than a multi-dimensional feature space (colour channels, spatial coordinates, texture features).

Because it requires no supervision and groups pixels purely by their statistical properties, thresholding qualifies as unsupervised segmentation.

Question 3.4 - Differences Between K-Means and Mean-Shift

Aspect	K-Means	Mean-Shift
Number of clusters	Must be specified a priori (*k*)	Determined automatically from data density
Cluster shape	Assumes spherical (Voronoi) clusters	Can discover arbitrarily shaped clusters
Initialization	Sensitive to initial centroid placement (mitigated by k-means++ / multiple inits)	No initialization needed; each point starts as a candidate
Key parameter	*k* (number of clusters)	*bandwidth* (kernel radius)
Scalability	$O(n \cdot k \cdot i)$ - scales well to large datasets	$O(n^2)$ per iteration - much

Aspect	K-Means	Mean-Shift
		slower on large data
Convergence	Converges to a local minimum of inertia	Converges to local modes of the density function
Outlier handling	Outliers are forced into the nearest cluster	Outliers can form their own small clusters

---

Question 3.5 - Limitations

K-Means limitations:

- Requires the number of clusters *k* to be chosen in advance
- Assumes clusters are convex and roughly equal in size
- Sensitive to initial centroid placement - can converge to poor local minima
- Only considers colour similarity; ignores spatial coherence unless location is added to the feature vector
- Not robust to noise or outliers

Mean-Shift limitations:

- Computationally expensive - $O(n^2)$ makes it impractical on high-resolution images without downsampling or bin seeding
- The bandwidth parameter heavily influences results and is not always easy to choose
- Struggles with high-dimensional feature spaces
- No guarantee on the number of clusters found - small bandwidth can over-segment, large bandwidth can under-segment
- Like K-Means, the basic form uses only colour features without spatial regularization unless location is explicitly included

Shared limitations:

- Neither method produces semantically meaningful segments - they group by appearance, not by object identity
- Both are sensitive to the colour space used (RGB vs. LAB vs. HSV can produce very different results)
- Neither inherently handles texture or shape information

Acknowledgment

I used a GPT-5.3-Codex to help scaffold and organize this notebook.

Gemini-3.1 was used to check the result and validate conformity with the assignment outline.

All observations and responses were written by me, Adam Nelson-Archer.